AI Diet Planner

A Project Report

Submitted in Partial Fulfilment of the Requirements
for the Award of the Degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE)

By

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Under the esteemed guidance of

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DEPARTMENT OF COMPUTER SCIENCE
ZSCT'S THAKUR SHYAMNARAYAN DEGREE COLLEGE
COLLEGE
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MUMBAI - 400101
MAHARASHTRA

2025-2026

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DEPARTMENT OF COMPUTER SCIENCE

CERTIFICATE

This is to certify that the project entitled "**AI Diet Planner**" is the Bonafede work of **Abhijit Tapas Bhakta** submitted in partial fulfillment of the requirements for the award of degree of **BACHELOR OF SCIENCE** in **COMPUTER SCIENCE** from THE University of Mumbai.

Internal Guide

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Introduction

The AI Career Coach Agent project is an intelligent and comprehensive solution designed to simplify and personalize the process of career development and job seeking. Traditionally, individuals face significant challenges in navigating their career paths, from building an effective resume to identifying the right skills for a desired role. This manual process is often fragmented, overwhelming, and lacks tailored, data-driven guidance, leading to uncertainty and stress during crucial career transitions.

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Architecture & Components:
Frontend:
Framework: Next.js, React, Tailwind CSS
Description: The entire user interface is a modern and responsive Single Page Application (SPA) built with React and the Next.js framework. This handles server-side rendering, routing, and API integration. All styling is implemented using Tailwind CSS, with pre-built components from Shad CN to accelerate UI development.

ABSTRACT

This project details the comprehensive design, development, and deployment of a full-stack **AI Diet Planner**, a sophisticated Software-as-a-Service (SaaS) web application. Built using a modern technology stack, the platform is engineered to provide users with a suite of intelligent tools aimed at enhancing their health and nutrition journey. The application's core architecture is centered around **Vite** for its fast development server and build capabilities, **React** for a dynamic and component-based user interface, and **Tailwind CSS** for rapid, utility-first styling.

1. **AI Nutrition Chatbot Agent:** This feature provides a real-time, conversational chat interface where users can ask complex nutrition-related questions. The agent leverages a Large Language Model (LLM) from the Groq API to act as an expert nutrition advisor, delivering structured, detailed, and relevant answers on topics ranging from meal suggestions to health tips.
2. **AI Health & Exercise Analyzer Agent:** Users have their health profile automatically analyzed by the system to generate personalized workout suggestions. The system calculates the user's BMI and sends their complete health profile to a dedicated AI agent. This agent evaluates the user's status (underweight, normal, overweight, or obese) against their goals and activity level, providing a detailed set of recommended exercises complete with duration, calorie burn estimates, and difficulty levels. The process includes a robust fallback to a static exercise database if the AI generation fails.

3. **AI-Powered Meal Plan Generator & Tracking:** This innovative tool generates personalized meal plans based on a user's dietary preferences, calorie targets, and health goals. Beyond generation, the platform offers comprehensive tracking for meals, weight, and daily nutrition, motivating users with a streak system that celebrates consistent logging and progress towards their target weight. This data is visualized through charts and a weekly overview calendar, offering clear and engaging feedback.

The technical architecture is robust and scalable, integrating several key services. User authentication and management are handled by **Supabase Auth**, which provides secure sign-up/sign-in flows (including Google OAuth) and protected routes. All user-generated data, including chat histories, meal logs, weight history, and user streaks, is persistently stored in a serverless **Supabase PostgreSQL** database.

A cornerstone of the backend is the direct integration between the frontend services and the **Supabase** and **Groq** APIs. The application's business logic, contained within a dedicated services layer, orchestrates the workflows for fetching data, logging meals, and calling the AI agents for meal generation and exercise recommendations. This Backend-as-a-Service (BaaS) approach ensures the application remains responsive and reliable by leveraging Supabase's managed infrastructure for data persistence and real-time capabilities.

ACKNOWLEDGEMENT

I am pleased to present this project, developing the “AI Diet Planner,” which represents an innovative step toward providing personalized, AI-driven nutrition and meal planning. The successful completion of this project would not have been possible without the invaluable support, encouragement, and guidance I received from many individuals along the way.

I would like to extend my heartfelt gratitude to our visionary mentor and our esteemed Principal, **Dr. G.D. Giri**, whose unwavering support and inspirational guidance laid the foundation for this project. Their expertise and dedication to the project's success have been a constant source of motivation.

I am deeply indebted to my Co-ordinator, **Mrs. Abha Dhote**, for her invaluable support, constant encouragement, and dedicated supervision throughout the course of this project. Her constructive feedback and motivation provided the right direction and helped me stay focused on achieving my goals.

I would also like to express my profound gratitude to my Project Guide, Mr. **Ajit sharma**, whose expertise, insightful guidance, and persistent encouragement were instrumental in shaping the AI Diet Planner. His valuable suggestions and timely mentorship played a crucial role in ensuring the successful development of this project.

I must express my sincere appreciation to my family and friends for their constant support, understanding, and encouragement throughout this project. Their belief in my abilities and unwavering support have been a source of strength.

This project represents a collective endeavor, and I am deeply thankful to all who have played a role in making it a reality. Your support, guidance, and expertise have contributed to the successful development of the **AI Diet Planner**, an innovative solution designed to offer personalized nutrition, meal planning, and health tracking tools. Thank you for being an integral part of this rewarding academic journey.

DECLARATION

I hereby declare that the project entitled, "**AI Diet Planner**" done at "**ZSCT's Thakur Shyamnarayan Degree College**", has not been in any case copied to submit to any other university for the award of any degree. To the best of my knowledge, no one has submitted it to any other university other than me.

The project is done in partial fulfilment of the requirements for the award of degree of **Bachelors of Science (Computer Science)** to be submitted as fifth semester project as part of our curriculum.

Name and Signature of the Student

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CHAPTER 1
INTRODUCTION

1.1 Introduction

The **AI Diet Planner** project is an intelligent and comprehensive solution designed to simplify and personalize the process of nutrition management and health tracking. Traditionally, individuals face significant challenges in navigating their dietary needs, from building an effective meal plan to identifying the right exercises for their fitness level. This manual process is often confusing, overwhelming, and lacks tailored, data-driven guidance, leading to inconsistency and frustration during crucial health and wellness journeys.

Project Overview:

The AI Diet Planner aims to resolve these challenges by providing a centralized, automated platform powered by artificial intelligence. Users can access a suite of specialized tools designed to offer instant, personalized support. The system integrates multiple AI agents that analyze user inputs—such as health profiles and dietary preferences—and generate actionable insights, structured meal plans, and expert-level nutritional advice, all within a single, interactive application. Each tool is carefully designed to provide clarity, boost motivation, and deliver a significant advantage in achieving personal health goals.

Key Highlights:

- **AI Nutrition Chatbot** for real-time, conversational answers to any nutrition-related questions.
- **AI Health & Exercise Analyzer** that provides instant, personalized exercise recommendations based on the user's BMI, activity level, and health goals.
- **AI-Powered Meal Plan Generator** to create dynamic, personalized meal plans based on dietary preferences and calorie targets.
- **Comprehensive Tracking & Gamification** to monitor daily nutrition, track weight progress, and build motivational logging streaks.
- **Secure Login and User Management** using Supabase Auth to maintain the confidentiality of personal health data and interaction history.
- **Personalized History Tracking** allowing users to save and revisit all their meal logs and weight progress reports.
- **Modern and Responsive Interface** built with React, Vite, and Tailwind CSS for a seamless experience across all devices.
- **Scalable and Reliable Architecture** using serverless technologies like Supabase and the Groq AI API to handle complex AI tasks efficiently.

More than just a tool, the AI Diet Planner is envisioned as a holistic health and wellness partner that empowers individuals, reduces the complexity of nutritional science, and democratizes access to personalized dietary guidance. It combines powerful automation with a user-friendly and flexible interface, making it adaptable for anyone looking to lose weight, gain muscle, or simply maintain a healthier lifestyle.

Ultimately, the AI Diet Planner aspires to create a lasting impact by transforming the traditional practice of diet planning into a dynamic, reliable, and personalized system. By streamlining this process, the platform not only improves efficiency and adherence but also ensures users receive consistent, high-quality, and data-driven support, empowering them to navigate their health journeys with confidence and clarity.

1.2 Objectives

The **AI Diet Planner** project is designed to empower individuals on their health and wellness journey by providing personalized, AI-driven tools while ensuring data privacy, scalability, and an intuitive user experience. The objectives include:

- **Personalized Nutrition Q&A** Provide an intelligent, real-time conversational agent capable of answering a wide range of complex nutrition-related questions, acting as an accessible, on-demand diet advisor.

- **AI-Powered Health Plan Generation** Create a tool that generates personalized meal plans based on user dietary preferences and goals, alongside tailored exercise recommendations that adapt to the user's specific BMI and activity level.

- **Comprehensive Health & Progress Tracking** Securely save all user-generated data—including meal logs, nutrition data, and weight history—to their personal account, allowing them to easily track progress and revisit past entries for continuous reference.

- **Reliable and Responsive AI Integration** Handle AI-powered tasks like meal and exercise generation efficiently through direct integration with the Groq API, ensuring the application remains responsive and provides a smooth user experience.|

- **Empower User Health and Motivation** Reduce the complexity and stress associated with diet planning by providing users with data-driven insights, structured plans, and instant feedback. Incorporate gamification elements like a daily logging streak system to boost engagement and consistency.

- **Secure and Centralized User Authentication** Provide a simple and robust authentication system using Supabase Auth that protects sensitive user data, including personal health profiles and dietary information, from unauthorized access.

- **Intuitive and Responsive User Interface** Deliver a modern, clean, and fully responsive user interface using React, Vite, and Tailwind CSS. The objective is to ensure a seamless, fast, and accessible user experience on both desktop and mobile devices, making health tracking tools easy to use for everyone.

1.3 Purpose, Scope, Architecture & Components:

1.3.1 Purpose

The primary purpose of the **AI Diet Planner** project is to democratize and personalize the health and nutrition management process, which is often fragmented, overwhelming, and lacks data-driven insights. By introducing AI-powered automation and a centralized platform, the system aims to bring clarity, confidence, and efficiency to individuals navigating their health and wellness journeys.

The specific purposes include:

- **Provide On-Demand, Expert Guidance** 💡 To eliminate the barriers to accessing professional nutrition advice by offering an intelligent platform that provides instant, expert-level answers and analysis 24/7, making guidance accessible to everyone.
- **Enhance Health & Fitness Outcomes** 🎯 To improve the quality of users' health choices by equipping them with personalized data, structured meal plans, and objective exercise recommendations, thereby reducing guesswork and enhancing strategic planning.
- **Create a Centralized Health Hub** 📂 To provide users with a single, unified platform to manage all aspects of their health planning—from asking questions and generating meal plans to logging meals, tracking weight, and monitoring their interaction history over time.

1.3.2 Scope

The **AI Diet Planner** project will focus on delivering a secure, scalable, and intuitive web application that provides a suite of AI-powered health and nutrition tools. The scope of the system includes:

- **Fully Responsive Web Application** 🖥️📱 The platform will be delivered as a modern web application, ensuring a seamless and accessible user experience across all devices, including desktops, tablets, and smartphones.
- **Core AI-Powered Agents** 🤖 The system will include distinct AI features as its core functionality: a Q&A Chatbot for real-time advice, an AI Health & Exercise Analyzer for personalized workout suggestions, and a Smart Meal Plan Generator for creating tailored meal plans.
- **User History and Data Management** 💾 The application will provide the ability for users to save, access, and manage their complete health data, including meal logs, weight tracking history, and daily logging streaks.
- **Secure User Authentication** 🔐 Each user must create an account and log in before accessing the personalized tools, maintaining the confidentiality and security of their personal health data using Supabase Authentication.
- **Responsive AI Task Processing** ⚙️ The system will handle AI-powered tasks, such as meal and exercise generation, efficiently through direct API integrations. This ensures the user interface remains fast, responsive, and is not blocked by long-running processes.

- **Interactive Data Visualization** 📊 Weight progress will be displayed using dynamic charts, and weekly nutrition data will be presented in a visual calendar, providing users with clear, engaging, and easy-to-understand representations of their health journey.

1.3.3 Architecture & Components:

Frontend:

- **Framework:** React, Vite, Tailwind CSS
- **Description:** The entire user interface is a modern and responsive Single Page Application (SPA) built with React and the Vite build tool. This handles all client-side rendering, routing, and API integration. All styling is implemented using Tailwind CSS, with pre-built components from shadcn/ui to accelerate UI development.

Backend:

- **Language:** TypeScript
- **Description:** The project utilizes a Backend-as-a-Service (BaaS) model, with core logic residing in the `src/services/` directory of the frontend application. These services, written in TypeScript, handle all business logic and orchestrate asynchronous calls to the Supabase backend for data operations and the Groq API for AI-powered features.

AI/ML Engine:

- **Engine:** Large Language Model (LLM) via Groq API
- **Description:** The application's intelligence is powered by the LLaMA 3 model, accessed through the Groq API. This engine is used to define and execute the specialized AI features, including the Nutrition Chatbot, the Meal Plan Generator, and the Exercise Recommendation agent.

Database:

- **Type:** Serverless Relational (PostgreSQL)
- **Technologies:** Supabase
- **Description:** The project uses the Supabase platform, which provides a serverless PostgreSQL database as its primary data store. It holds all user information, health profiles, meal logs, weight history, and engagement data like streaks. All database interactions are handled via the Supabase client library.

Background Worker:

- **Type:** Asynchronous API Calls

- **Description:**The application does not use a dedicated background worker engine. Instead, it leverages a BaaS architecture where potentially long-running tasks, like generating AI recommendations or meal plans, are handled as standard asynchronous API calls to the Groq and Supabase services. This ensures the UI remains fast and responsive for the user without being blocked.

Storage:

- **Primary:** Not Applicable
- **Description:** This application does not require dedicated cloud storage for user-uploaded files. All user-generated content, such as meal and weight logs, is stored as structured data directly in the Supabase PostgreSQL database.
-

Auth:

- **Type:** Managed Authentication Service
- **Description:** User authentication, authorization, and session management are fully handled by **Supabase Auth**. It provides secure sign-up, sign-in (with social providers like Google OAuth), and user profile management out-of-the-box, protecting user data and application routes through Row Level Security (RLS) policies.

Data Visualization:

- **Formats:** Interactive Charts, Calendar View, Custom Components
- **Technologies:**Recharts
- **Description:** The application's output is rendered dynamically in the browser. Historical weight data is visualized as interactive line charts using the **Recharts** library. AI-generated responses and exercise recommendations are displayed using custom React components, and the weekly nutrition summary is presented in a calendar-style view for easy tracking.

1.3.4 Summary

The **AI Diet Planner** is an intelligent, web-based SaaS platform that ensures clarity, personalization, and efficiency in health and nutrition management. By automating the often confusing and overwhelming process of diet planning and fitness tracking, it guarantees users receive data-driven insights, structured guidance, and tailored feedback. With features such as a suite of AI-powered agents for nutrition Q&A, health and exercise analysis, and meal plan generation; secure user authentication; personalized history management; motivational tools like a daily logging streak system; and interactive data visualization, the AI Diet Planner reshapes how individuals approach their health and wellness, ultimately empowering them with the confidence and clarity needed to achieve their health goals.

CHAPTER 2
SURVEY OF TECHNOLOGY

2.1 Problem definition

In today's fast-paced world, individuals face significant hurdles in managing their health and nutrition effectively. The traditional methods of dieting, meal planning, and exercise are often generic, difficult to sustain, and lack personalized, data-driven insights. This leads to inconsistency, frustration, and the failure to achieve long-term health goals.

Some of the key challenges include:

- **Lack of Personalized Guidance** 🧭 Generic diet plans and workout routines found online rarely address an individual's unique body composition, lifestyle, and health goals. Access to professional nutritionists and personal trainers is often expensive and inaccessible for many.
- **Meal Planning & Tracking Struggles** 📄 Creating balanced, nutritious meal plans that align with specific calorie and macronutrient targets is a major challenge. Individuals often struggle with the tedious process of accurately and consistently tracking their daily intake.
- **Unclear Fitness & Health Paths** 🛤️ Individuals looking to lose weight, gain muscle, or improve their health frequently struggle to find a clear, step-by-step roadmap for effective exercise and dietary changes. This ambiguity can delay or derail their progress.
- **Information Overload** 🤯 The vast amount of conflicting nutritional advice, fad diets, and workout trends can be paralyzing. It is difficult for individuals to sift through the noise and find reliable, science-backed information.
- **Fragmented Tooling** 🧩 People are forced to use a multitude of disconnected apps and tools—calorie counters, workout loggers, recipe finders, and progress trackers—with no single, integrated hub to manage their health and wellness activities.

The **AI Diet Planner** addresses these challenges by offering a centralized, intelligent platform that provides personalized guidance. It ensures users receive instant, data-driven feedback through exercise analysis, clear and dynamic meal plans, and on-demand answers to their nutrition questions, thereby streamlining their health journey and empowering them with confidence.

2.2 System Requirements

The following hardware and software requirements must be met to successfully develop and run the **AI Career Coach Agent** application:

2.2.1 Hardware Requirements

- **Processor:** Intel Core i5 or higher (or equivalent AMD processor)
- **RAM:** Minimum 8GB (16GB recommended for a smoother development experience)
- **Storage:** At least 256GB SSD for fast code compilation and dependency installation
- **Network:** A stable, high-speed internet connection is essential for accessing cloud services, APIs, and deploying the application.

2.2.2 Software Requirements

- **Programming Language:** TypeScript
- **Frameworks & Tools:** React, Vite
- **Database:** PostgreSQL (provided via Supabase)
- **Authentication:** Supabase Auth
- **Backend Platform:** Supabase
- **AI / LLM API:** Groq API
- **Development Environment:** Node.js (v18.x or higher), a code editor like VS Code
- **Deployment Platform:** Vercel (recommended) or a similar modern cloud hosting provider

2.3 Justification of Selection of Technology

- **Frameworks & Build Tool (React & Vite)** ⚛️ This combination was chosen to build a modern, high-performance, and responsive web application. React provides a powerful component-based architecture for creating dynamic and interactive user interfaces. Vite was selected for its extremely fast development server and optimized build process, which significantly accelerates the development lifecycle and improves overall application performance, making it an ideal choice for a modern Single Page Application (SPA).

- **Backend as a Service (Supabase)** ☁️ Supabase was selected as the all-in-one backend platform because it provides a serverless PostgreSQL database, managed authentication (including social providers), and Row Level Security out-of-the-box. This BaaS (Backend-as-a-Service) approach drastically reduces backend development time and complexity, allowing the focus to remain on core application features like AI integration and user experience. Its scalability and built-in security features make it a robust and secure choice for managing sensitive user health data.

- **AI Engine (Groq API)** 🤖 Groq was chosen for its high-speed AI inference capabilities, which are essential for providing a real-time, conversational experience in the AI chatbot and for generating meal plans and exercise recommendations quickly. Its availability of a free tier and simple API integration make it an excellent choice for powering the application's core intelligence features without incurring significant initial costs, ensuring a responsive and engaging user experience.

- **UI Components & Styling (shadcn/ui & Tailwind CSS)** 🎨 This combination was chosen for rapid and consistent UI development. Tailwind CSS provides a utility-first framework that allows for building custom, responsive designs directly in the markup without writing traditional CSS files. Shadcn/ui was selected because it offers a library of beautifully designed, accessible, and customizable components built on top of Tailwind, significantly accelerating the development of a polished, modern, and professional-looking user interface.

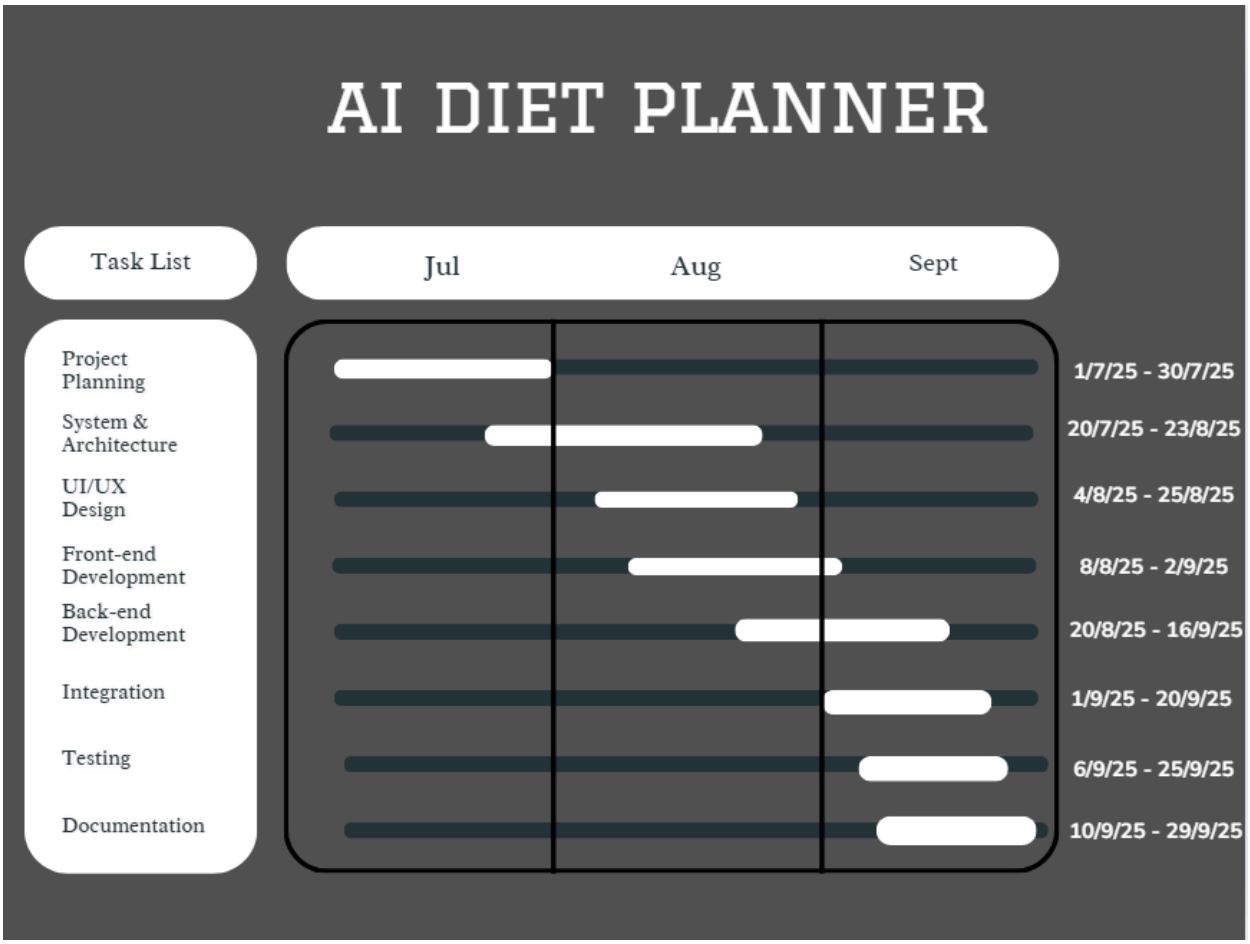
2.4 Planning and Scheduling

The development of the **AI Diet Planner** follows a systematic and phased approach to ensure a robust architecture, seamless integration of AI services, and timely delivery of a high-quality SaaS platform. The planned stages are outlined below:

1. **Requirement Analysis & Research**
    - Gather detailed requirements based on the needs of individuals seeking to manage their health, track nutrition, and achieve fitness goals.
    - Define the functional requirements for each AI feature: the Nutrition Chatbot, the Meal Plan Generator, and the Exercise Analyzer.
    - Finalize the scope and objectives of the application, focusing on personalization, tracking, and motivation.
2. **System & Database Design**
    - Design the database schema for storing user data, including user_profiles, user_health_profiles, meal_logs, weight_logs, and user_streaks.
    - Define the overall system architecture, outlining the roles and interactions between the React/Vite frontend, the Supabase backend (database and auth), and the Groq API for AI.
    - Plan the data flows for the AI-driven features, such as sending user health profiles to the Groq API to generate exercise recommendations.
3. **Backend Development & AI Integration**
    - Implement the core application logic within the client-side services layer (src/services/) to interact with the BaaS provider.
    - Develop and configure the AI features by integrating with the Groq LLM API to power the chatbot, meal generator, and exercise analyzer.
    - Integrate Supabase Auth for secure user authentication and session management.
    - Build the data access layer using the Supabase client library to communicate with the PostgreSQL database.

4. **Frontend Development**
    - Build the user-friendly interfaces for the dashboard, chat agent, meal and weight logging dialogs, and the onboarding flow using React and Vite.
    - Implement the interactive data visualizations, such as weight progress charts using the Recharts library.
    - Ensure a fully responsive and accessible design for all devices using Tailwind CSS and shadcn/ui components.
5. **Integration**
    - Connect the frontend components with the backend services to enable seamless data exchange with Supabase and the Groq API.
    - Ensure the asynchronous AI generation workflows correctly update the user interface upon completion (e.g., displaying exercises after a loading state).
    - Integrate the Supabase authentication flows and protect application routes based on user session status.
6. **Testing**
    - Conduct unit testing for individual React components and service functions.
    - Perform integration testing to validate that all services (Frontend, Supabase DB/Auth, Groq API) work together correctly.
    - Carry out end-to-end user acceptance testing on all major workflows, including user onboarding, logging a meal, and generating an exercise plan.
    - Test for UI responsiveness, cross-browser compatibility, and performance.
7. **Deployment**
    - Configure the production environment on Vercel, including all necessary environment variables for Supabase and Groq.
    - Deploy the Vite/React application and connect it to the production Supabase project.
    - Set up a custom domain, SSL certificate, and ongoing monitoring for the live application.

2.4.1 GANTT Chart

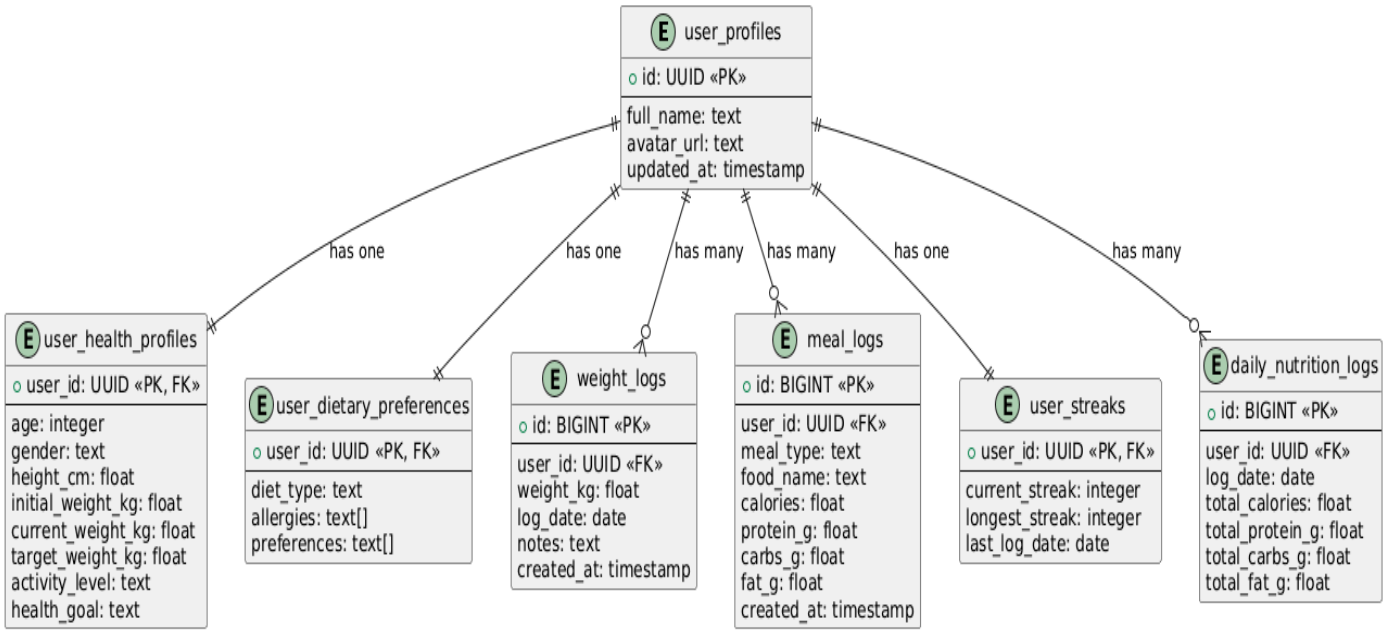


CHAPTER 3
SYSTEM DESIGN

3.1 UML Diagrams

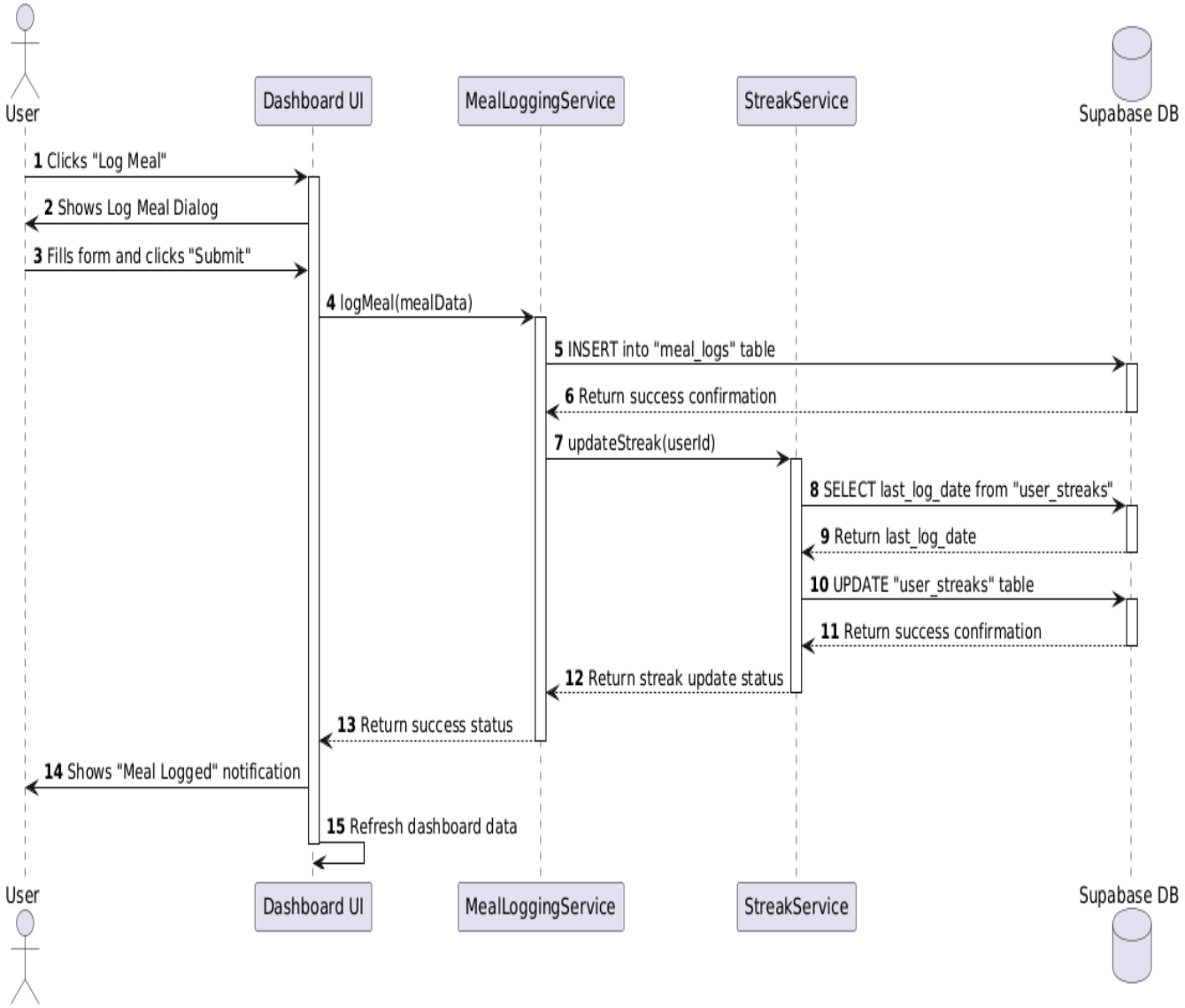
3.1.1 ER Diagram

An Entity-Relationship (ER) Diagram is a visual representation used in database design and modeling. It describes the relationships between various entities or tables in a database and the attributes associated with them. ER diagrams use symbols and lines to illustrate how different entities relate to each other, helping to define the structure of a database and how data is organized within it. These diagrams are essential for database planning, design, and communication between stakeholders involved in a database project.



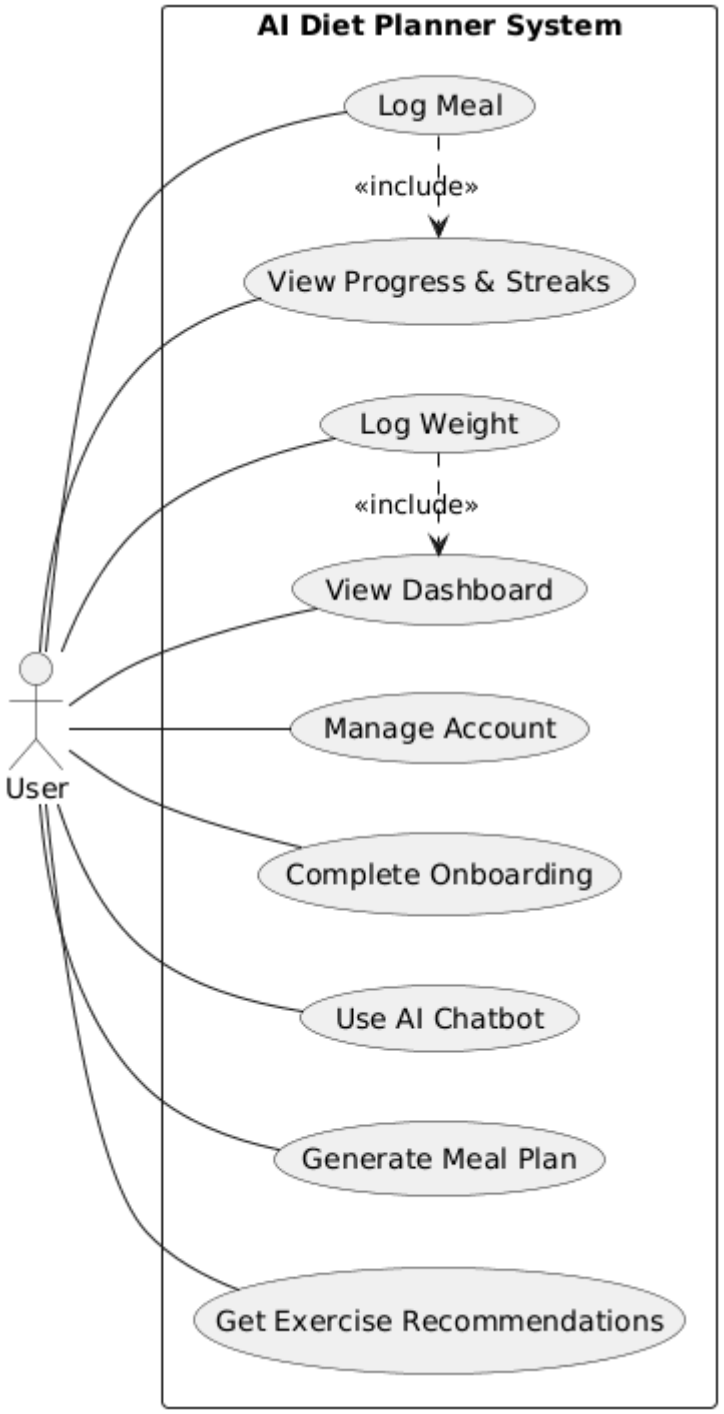
3.1.2 Sequence Diagram

A Sequence Diagram is a type of interaction diagram in Unified Modeling Language (UML) that visualizes the interactions between different objects or components in a system over time. It focuses on the chronological order of messages or interactions between these objects, showing how they collaborate to accomplish a specific task or scenario. Sequence diagrams are used for system design, analysis, and documentation, providing a clear representation of the dynamic behavior and flow of messages within a system or during a particular operation or use case.



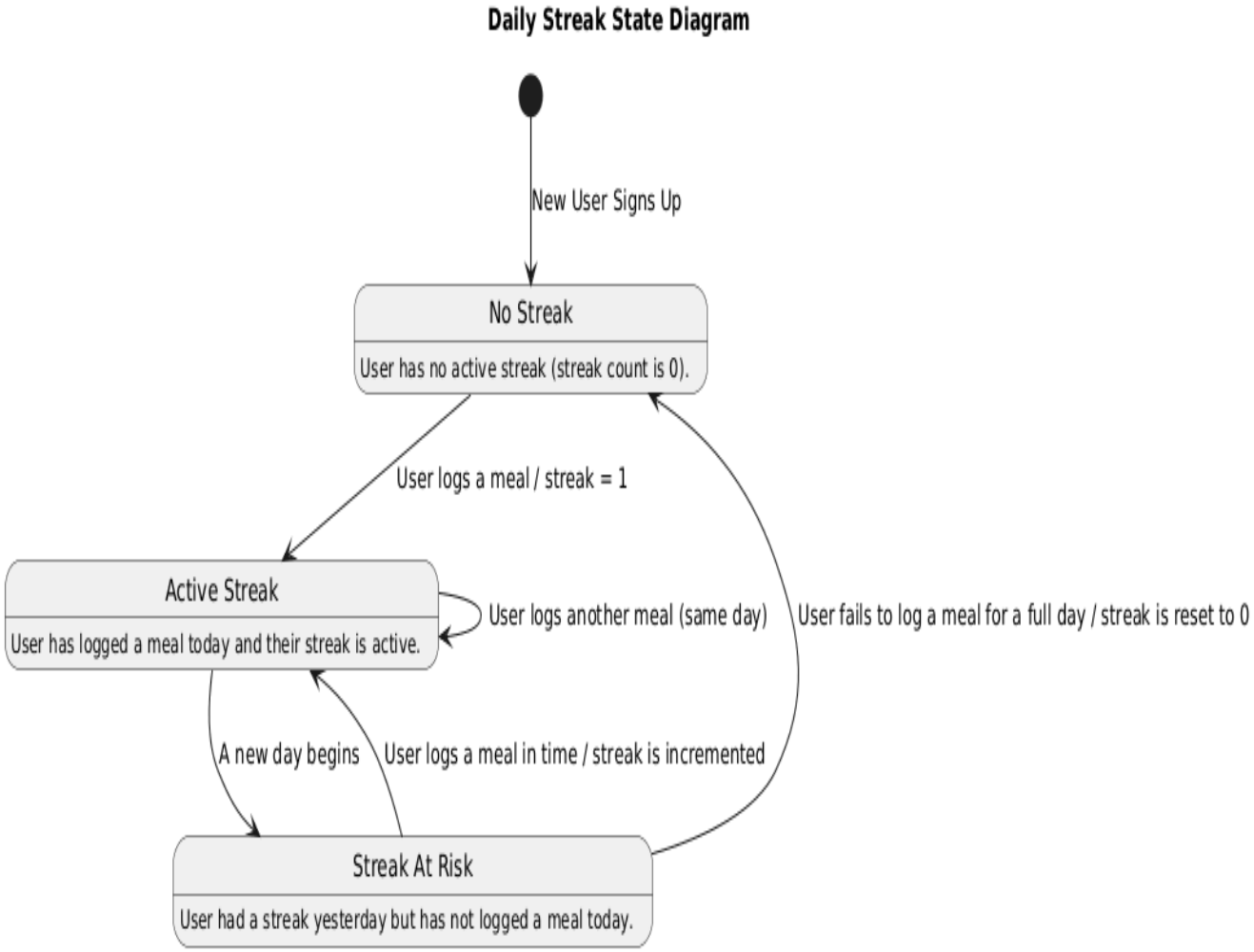
3.1.3 Use Case Diagram

A Use Case Diagram is a visual representation in Unified Modeling Language (UML) that depicts how users or external entities interact with a system, highlighting the specific functionalities, or "use cases," that the system provides in response to those interactions It's a useful tool for requirements analysis, design, and documentation in software development.



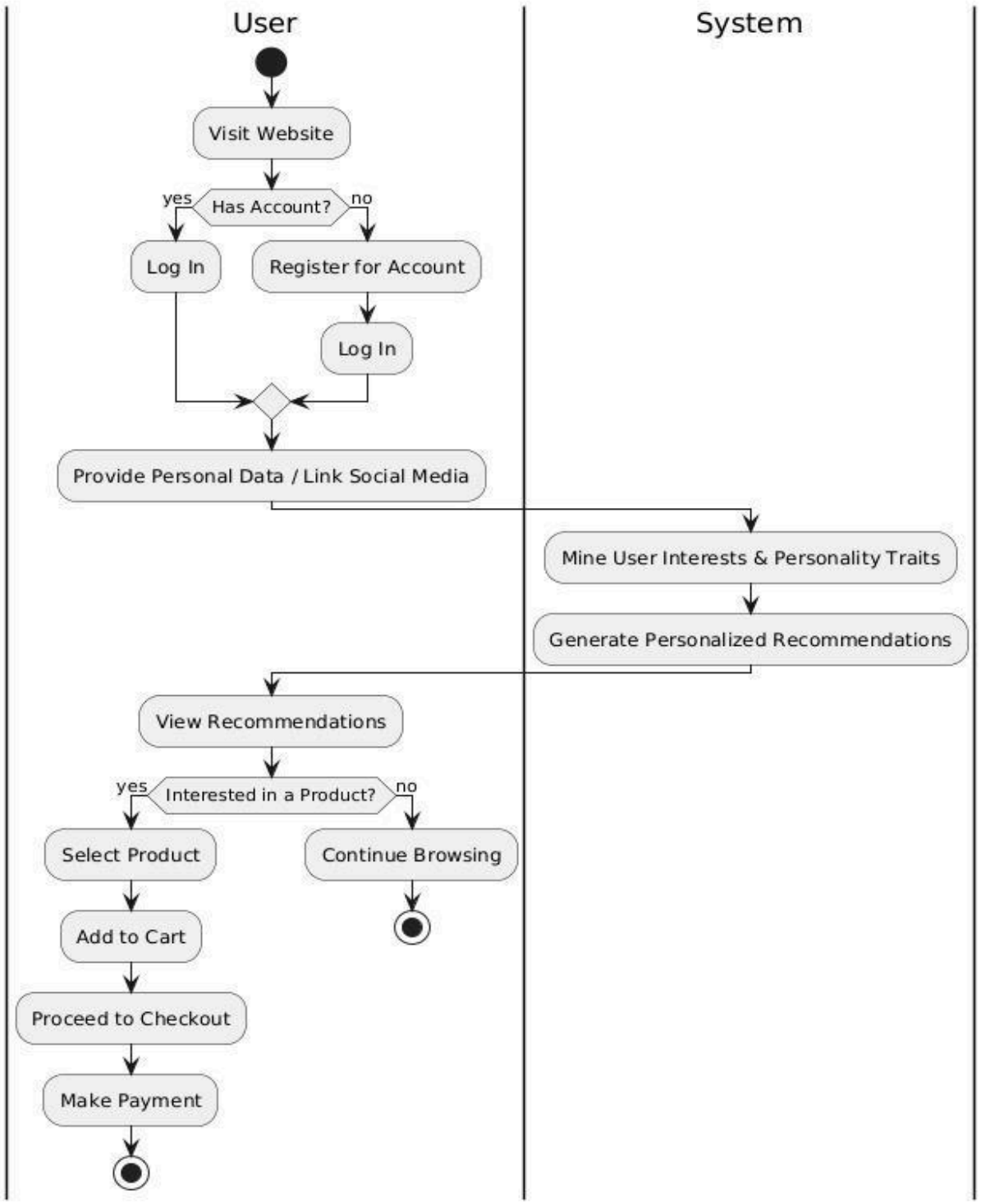
3.1.4 State Diagram

A state diagram, also known as a state machine diagram, is a visual representation in UML (Unified Modeling Language) that depicts the various states an object or system can be in and the transitions between these states. These diagrams are used to model the dynamic behavior of a system, showing how it responds to various events, conditions, or inputs by transitioning between different states. State diagrams are beneficial for designing and understanding the logic and control flow within a system, making them valuable tools in software engineering and other fields like electrical engineering and control systems.



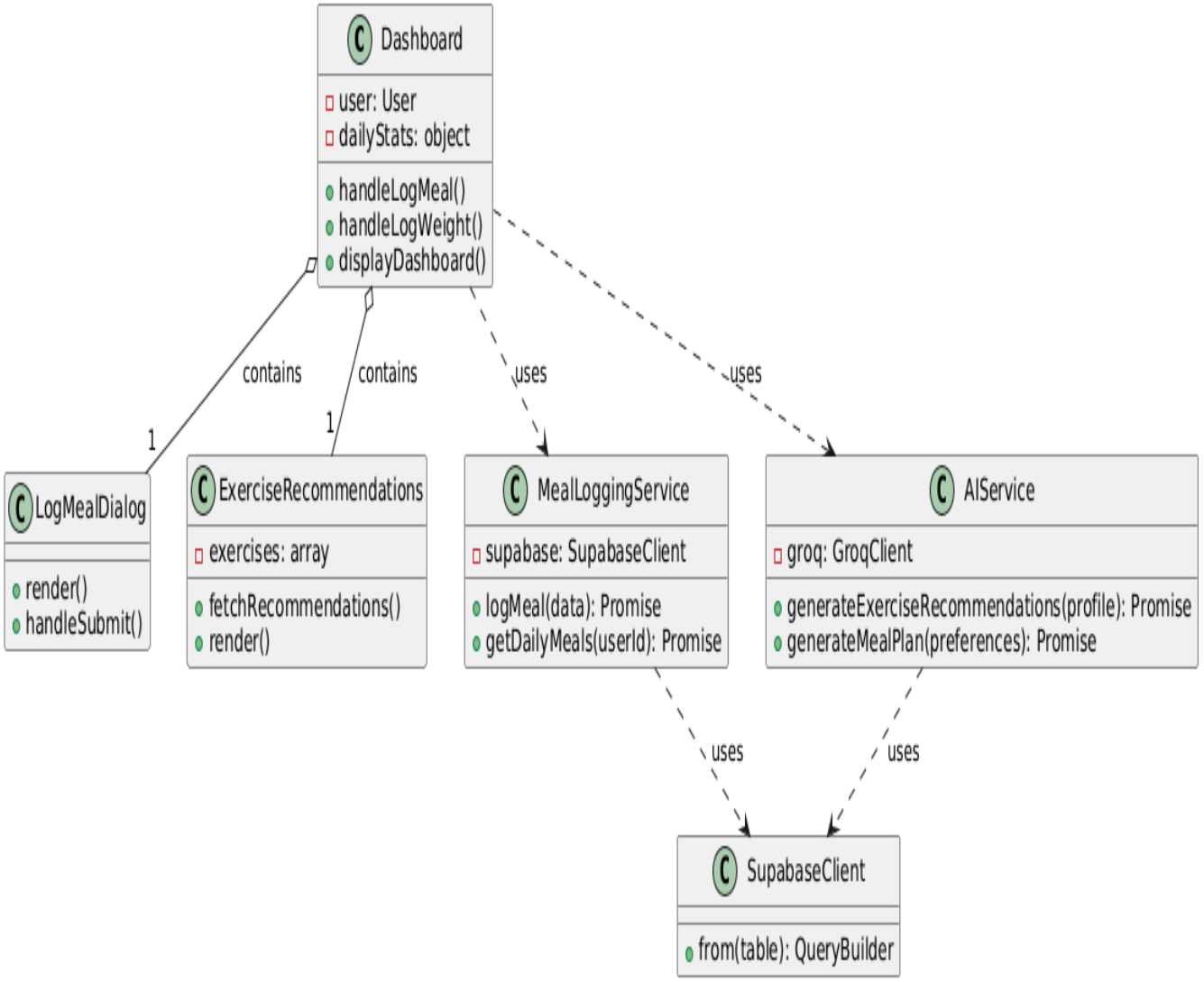
3.1.5 Activity Diagram

An activity diagram, in the context of UML (Unified Modeling Language), is a visual representation used to model the dynamic aspects of a system, focusing on the flow of activities or actions within a business process, software application, or any complex system. It is particularly useful for illustrating how various activities are organized, executed, and interact with each other, providing a clear, high-level view of processes, workflows, and the order of tasks in a system.



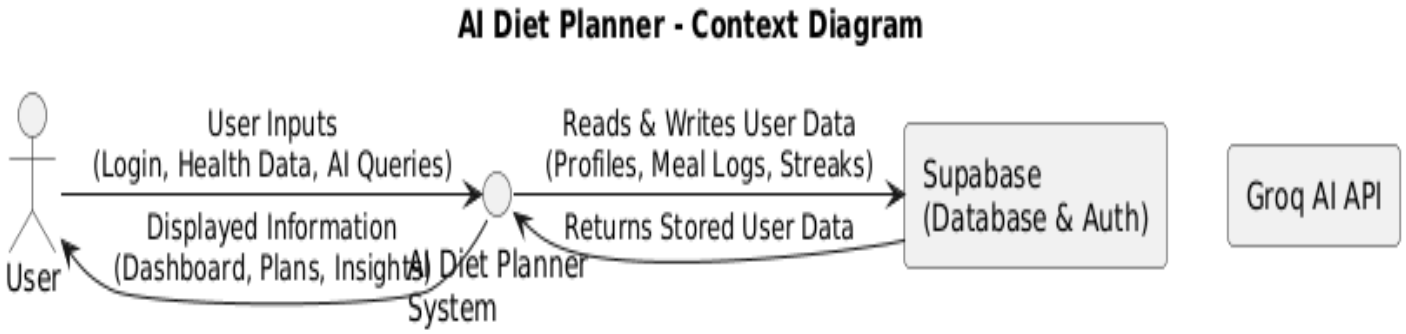
3.1.6 Class Diagram

A class diagram, illustrates the classes or objects in the system, their attributes (data), methods (functions), and the relationships between them. Class diagrams are a fundamental tool in software engineering and serve as a blueprint for understanding and designing the architecture of object-oriented systems, making them essential for both developers and stakeholders to visualize the structure and behavior of a software application.



3.1.7 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation used in systems engineering and software design to depict how data moves through a system. It illustrates processes (functions), data stores, data flows, and external entities and their interactions. DFDs are particularly useful for understanding the flow of data within a system or between systems, making them a valuable tool for analyzing, designing, and documenting information systems and business processes.



CHAPTER 4
IMPLEMENTATION & TESTING

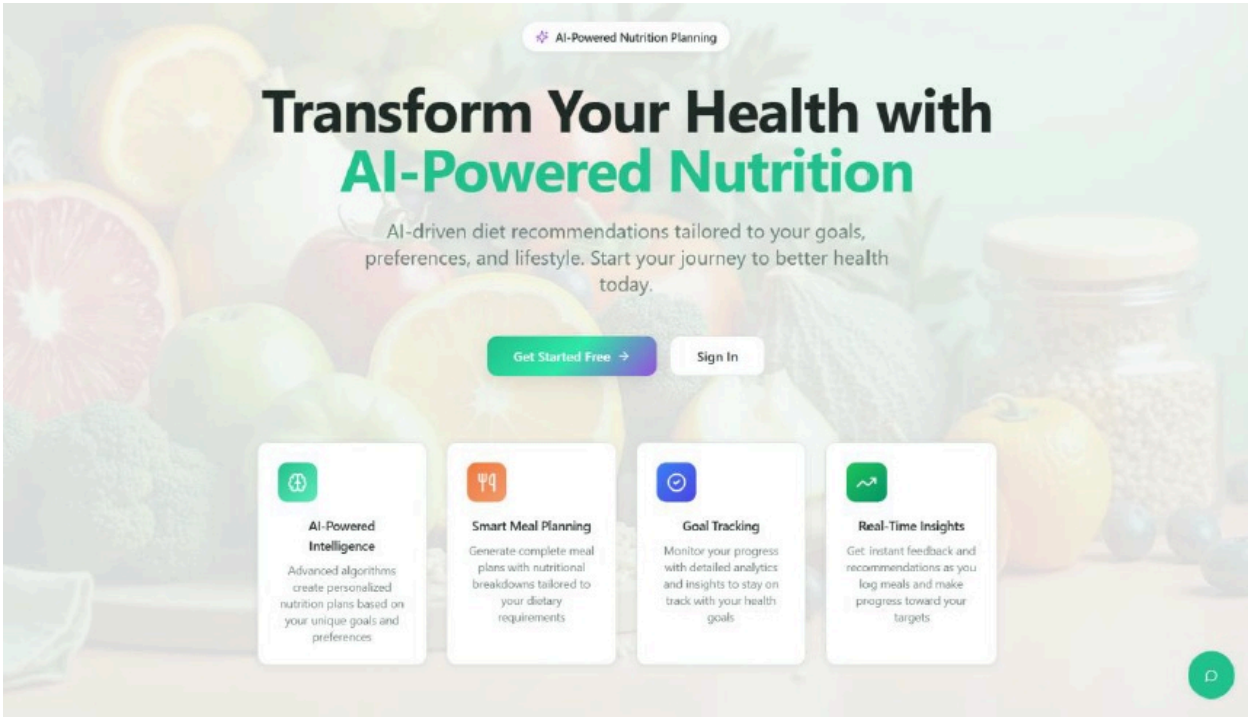
4.1 Test Case design/Test Case Scenario

Test cases help guide the tester through a sequence of steps to validate whether a software application is free of bugs, and working as required by the end-user. Typically, test cases for a given module or part of an application, are grouped into a test suite. More often than not, a test session will include many test cases because there will usually be more than one specific scenario to be tested.

Test Case 1: Landing Page

Image Reference: image_062fe6.jpg

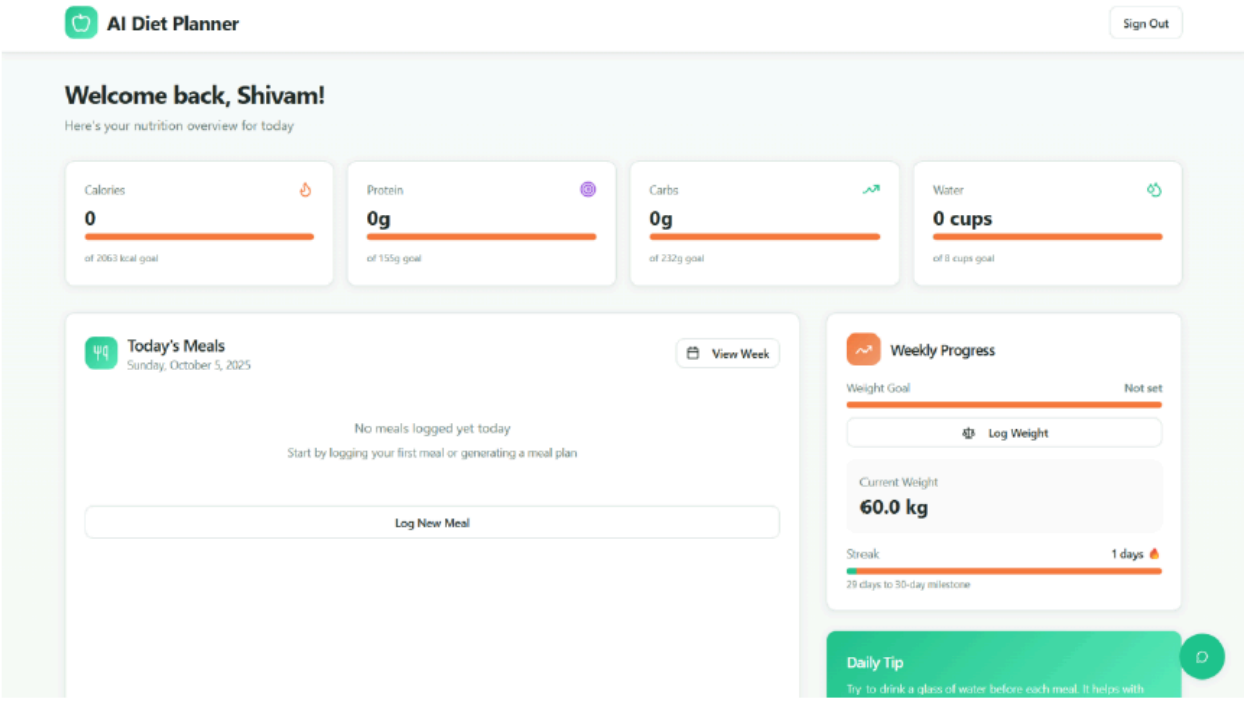
- **Test Case ID:** TC-LANDING-001
- **Test Case Title:** Verify "Get Started Free" button navigation for new users.
- **Preconditions:** The user is a new visitor and is not logged into the application.
- **Steps:**
    1. Navigate to the application's main landing page.
    2. Locate the primary call-to-action button labeled "**Get Started Free ->**".
    3. Click the "**Get Started Free ->**" button.
- **Expected Result:** The user is successfully redirected to the sign-up/registration page to create a new account.
- **Status:** Pass



Test Case 2: Main Dashboard

Image Reference: image_063021.png

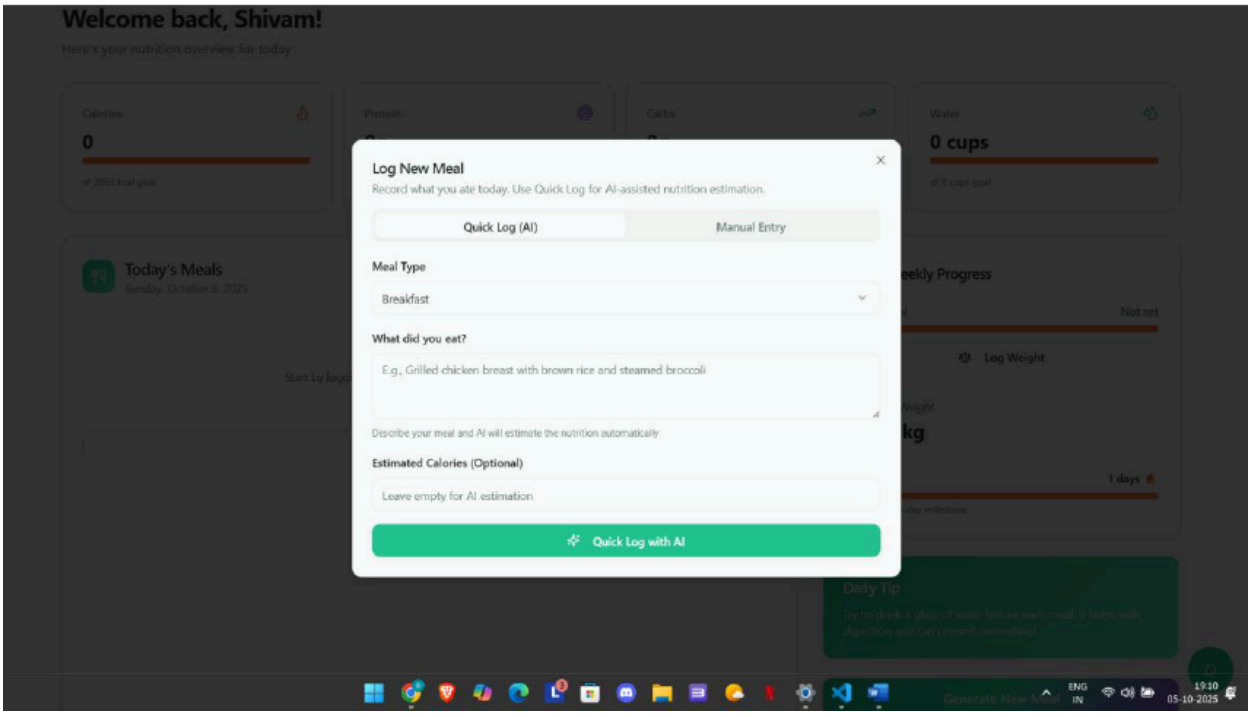
- **Test Case ID:** TC-DASH-001
- **Test Case Title:** Verify dashboard widgets display correct initial data upon login.
- **Preconditions:** The user "Shivam" is an existing user who is logged in. The user's current weight is 60.0 kg, and their current logging streak is 1 day. No meals have been logged for the current day (Sunday, October 5, 2025).
- **Steps:**
    - The user successfully logs in and is directed to the main dashboard.
    - Observe the data in the "Weekly Progress" and "Today's Meals" cards.
- **Expected Result:**
    - The "Today's Meals" card correctly displays "No meals logged yet today".
    - The "Weekly Progress" card correctly displays the "Current weight" as **60.0 kg**.
    - The "Weekly Progress" card correctly displays the "Streak" as **1 day**.
- **Status:** Pass



Test Case 3: Log New Meal (AI)

Image Reference: image_063060.png

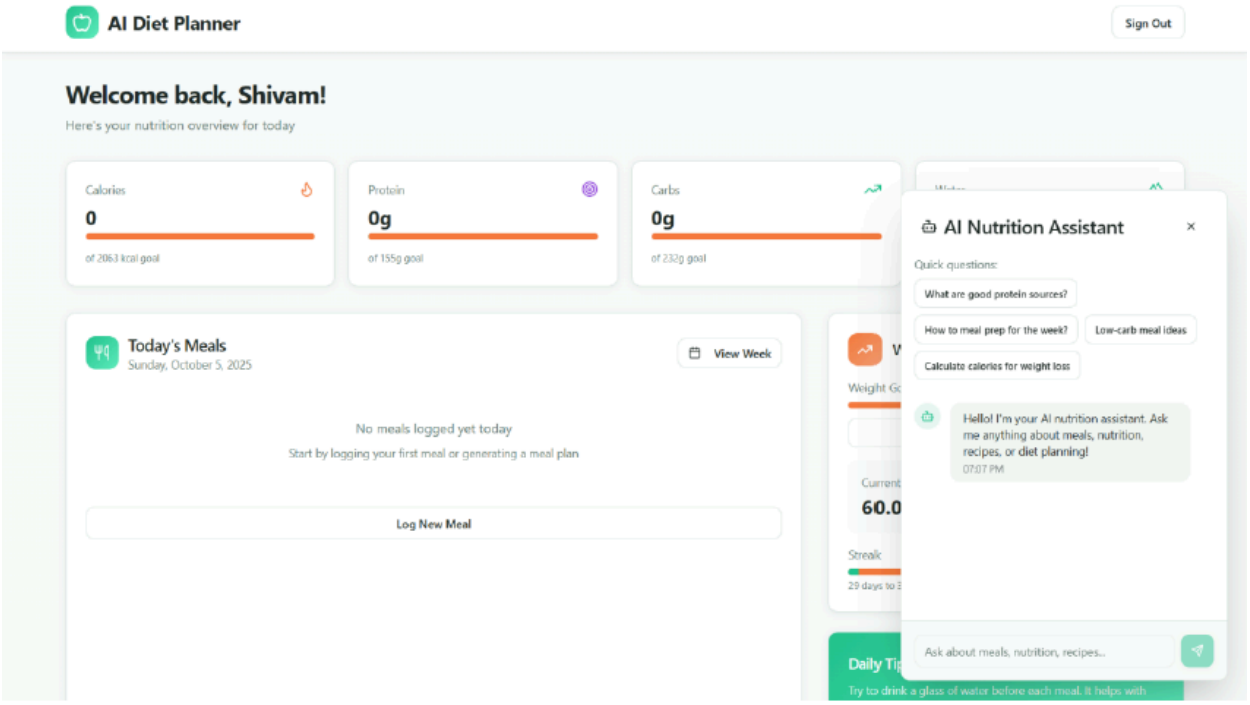
- **Test Case ID:** TC-MEAL-AI-001
- **Test Case Title:** Verify successful meal logging using the "Quick Log (AI)" feature.
- **Preconditions:** The user is logged in and on the main dashboard.
- **Steps:**
    1. Click the "**Log New Meal**" button.
    2. In the "Log New Meal" dialog, ensure the "**Quick Log (AI)**" tab is selected.
    3. Select a "Meal Type" from the dropdown (e.g., 'Lunch').
    4. Enter a natural language description in the "What did you eat?" text field, such as "**Grilled chicken breast with brown rice and steamed broccoli**".
    5. Click the "**Quick Log with AI**" button.
- **Expected Result:** The dialog closes, a success notification appears, and the main dashboard updates. The "Calories," "Protein," and "Carbs" widgets now show values greater than zero, reflecting the AI's estimation of the meal.
- **Status:** Pass



Test Case 4: AI Nutrition Assistant

Image Reference: image_05bb65.png

- **Test Case ID:** TC-CHAT-001
- **Test Case Title:** Verify the AI chatbot responds correctly to a "Quick question" button.
- **Preconditions:** The user is logged in and the "AI Nutrition Assistant" chat window is open.
- **Steps:**
    1. Locate the "Quick questions" area in the chat window.
    2. Click the button labeled "**What are good protein sources?**".
- **Expected Result:** The user's query ("What are good protein sources?") appears in the chat history, followed by a relevant, helpful response from the AI assistant that lists various sources of protein (e.g., chicken, fish, beans, tofu).
- **Status:** Pass



4.2 Testing Approach

Of course. Here is a comprehensive testing approach for the **AI Diet Planner** project, tailored to its specific features and architecture, following the format you provided.

Testing Approach

The objective of this testing phase is to ensure the AI Diet Planner application is robust, reliable, and provides an intuitive user experience. The approach combines functional, usability, and error handling tests to validate every aspect of the system, from user authentication to the performance of the AI features. 🧪

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1. Functional Testing

This is the primary testing effort, focused on verifying that each feature and module operates according to its specifications.

Authentication Module (Supabase Auth)

This tests the entire user lifecycle, from sign-up to login and session management.

- **Successful Sign-Up & Login (Happy Path):** Verify that a new user can sign up successfully using Google or email/password. Subsequently, confirm the user can log in with the created credentials and is redirected to the onboarding flow or the main dashboard.
- **Failed Login (Negative Paths):**
    - **Invalid Credentials:** Attempt to log in with an incorrect password or a non-existent email address. The interface should display the appropriate error message from Supabase.
    - **Access Control:** Attempt to access a protected route (e.g., the dashboard) without being logged in. The application should redirect the user to the sign-in page.

AI Nutrition Chatbot

This tests the core interactive chat functionality.

- **Successful Interaction (Happy Path):**
    - Open the chat and ask a relevant nutrition question (e.g., "What are some healthy snack ideas?").
    - Verify that the user's message appears instantly, a loading indicator is shown, and a coherent, well-formatted response from the AI is displayed.
- **Error Handling:**
    - Simulate a failure or timeout from the Groq API. The frontend should stop the loading indicator and display a user-friendly error message (e.g., "Sorry, the assistant could not respond. Please try again.").

Meal & Weight Logging Module

This tests the core data entry workflows for tracking user progress.

- **Successful Logging (Happy Path):**
  - **AI Meal Log:** Use the "Quick Log (AI)" feature to enter a meal description (e.g., "A bowl of oatmeal with berries and nuts"). Verify that the dialog closes, a success notification appears, and the dashboard's nutrition widgets (Calories, Protein, etc.) update with AI-estimated values.
  - **Weight Log:** Open the "Log Weight" dialog, enter a valid weight, and submit. Verify the dialog closes and the "Current Weight" widget on the dashboard updates to the new value.
- **Input Validation & Error Handling:**
  - **Invalid Meal Input:** Attempt to submit a meal log with an empty description. The "Quick Log with AI" button should be disabled or show a validation error.
  - **Invalid Weight Input:** Attempt to submit a non-numeric or negative value for weight. The form should display an appropriate error message.

AI Exercise Recommendation & Streak System

This verifies the automated recommendation and gamification logic.

- **Successful Generation (Happy Path):**
  - For a user with a completed health profile, log in and navigate to the dashboard.
  - Verify that a set of personalized exercise recommendations are displayed, with each card containing a title, description, duration, and calorie burn estimate.
- **Streak Logic Persistence:**
  - Log a meal for the first time. Verify the streak on the dashboard updates to "1 day".
  - Return the next day and log another meal. Verify the streak increments to "2 days".
  - Skip a full day without logging. Return on the following day and log a meal. Verify the streak resets to "1 day".
- **AI Failure (Fallback):**
  - Simulate a failure in the Groq API call for exercises. Verify that the system gracefully loads a set of pre-defined static exercises instead of showing an error, ensuring the user always receives guidance.

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2. UI/UX (Usability) Testing

This focuses on ensuring the application is intuitive, visually appealing, and easy to use. 🖥️

- **Responsiveness:**
  - Resize the browser window to simulate desktop, tablet, and mobile screen sizes.
  - Verify that all components—dashboard cards, dialogs, and the chat interface—adjust their layout correctly without content overlapping or becoming unusable.
- **Visual Consistency & Feedback:**
  - Navigate through all parts of the application and check for consistent styling

(fonts, colors, button styles) as defined by the shadcn/ui and Tailwind CSS framework.
    - Confirm that loading states (skeletons, spinners on buttons) are clearly visible during data fetching or AI processing.
    - Ensure all interactive elements (buttons, links, input fields) have clear hover and focus states.

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3. Test Environment and Data

- **Test Environment:**
    - **Browsers:** Google Chrome, Mozilla Firefox, and Apple Safari (latest versions).
    - **Backend Services:** A staging or development environment on Vercel connected to a test **Supabase** project and a valid **Groq API** key.
- **Test Data:**
    - **Supabase User Accounts:** At least three test accounts with different, complete health profiles: one targeting weight loss (BMI > 25), one targeting weight gain (BMI < 18.5), and one with a normal BMI.
    - **Sample Queries:** A list of standard nutrition questions for the chatbot (e.g., "Is coffee healthy?", "How much water should I drink?") and various meal descriptions for the AI logger (e.g., "2 eggs and toast", "chicken salad sandwich").

CHAPTER 5
RESULT & DISCUSSION

Result

The development and implementation of the AI Diet Planner successfully met its objectives by providing a reliable and efficient full-stack platform for AI-driven health and nutrition guidance. 🚀 The system is capable of generating personalized, structured, and actionable health insights in real-time, leveraging a suite of powerful AI features.

Features such as a secure user authentication system (via Supabase Auth), an interactive AI Nutrition Chatbot, an in-depth AI Health & Exercise Analyzer, and a Smart Meal Plan Generator, alongside comprehensive tracking for meals, weight, and motivational streaks, were tested and verified to function seamlessly. The integration of Supabase as an all-in-one backend with its PostgreSQL database for persistent user history ensures the platform is both responsive and robust. Overall, the project delivers a cohesive and valuable tool that empowers users in their health and wellness journey.

Discussion

The results clearly demonstrate that the AI Diet Planner significantly empowers users with immediate, on-demand access to personalized health and nutrition guidance, reducing the manual research and uncertainty involved in diet and fitness planning. 🥗 The platform's core AI features, including the AI Health & Exercise Analyzer and Smart Meal Plan Generator, replace generic advice with structured, data-driven insights by providing objective feedback and clear, actionable health plans.

Usability is greatly enhanced by the persistent history and gamification features, which allow users to track their progress by revisiting past meal logs and weight entries, and build motivation through a daily streak system, creating a continuous health and wellness log. This is all protected by a simple and secure authentication system via Supabase Auth, which safeguards sensitive personal data like health profiles and meal history, ensuring user privacy and building trust in the platform.

CHAPTER 6
CONCLUSION

6.1 Conclusion

The **AI Diet Planner** demonstrates how thoughtful full-stack design and intelligent automation can significantly simplify the complex process of health and nutrition management for individuals. 💡 By combining powerful, AI-driven features with a user-centric interface, it empowers users with personalized and actionable insights, eliminating their reliance on generic advice and the repetitive task of sifting through overwhelming information. The project's modern, services-based architecture ensures scalability, reliability, and a seamless user experience.

Overall, the **AI Diet Planner** delivers a balance of efficiency, personalization, and data-driven guidance. It transforms health management from a daunting task into an insightful, secure, and reliable process—ultimately empowering users with the tools and confidence needed to navigate their health and wellness journey effectively while upholding the privacy of their personal data

6.2 Future Scope

Looking ahead, the **AI Diet Planner** can be enhanced further to provide deeper, more proactive health guidance and lifestyle management tools. Some of the future directions include:

1. AI-Powered Nutritional Gap Analysis

This feature would bridge the gap between a user's current dietary habits and their ideal nutritional intake. 🔬

- Allow users to log their meals over a set period (e.g., one week).
- The system will automatically analyze the user's logged meals against the specific macronutrient and micronutrient targets recommended for their health goals.
- Generate a detailed **Weekly Nutrition Report**, clearly highlighting nutritional deficiencies (e.g., "low in fiber," "needs more protein") and excesses (e.g., "high in sodium").
- Provide a personalized and actionable plan with specific food suggestions or meal adjustments to help the user correct these identified gaps.

2. Interactive Workout Plan Builder & Tracking

This would transform the platform from providing recommendations into a comprehensive fitness management tool. 💪

- Integrate a new module for building and tracking multi-week workout programs.
- Based on the user's profile, goals, and available equipment, the AI agent will generate a structured, week-by-week workout plan.
- The module would allow users to check off completed exercises, log sets/reps/weights, and monitor their performance over time.
- The AI would then provide feedback on consistency and could dynamically adjust the following week's plan based on the user's logged performance and feedback (e.g., "increase weight on squats," "reduce cardio duration to focus on recovery").

CHAPTER 7
REFERENCES & GLOSSARY

References

1. **Vite:** https://vitejs.dev/
2. **React:** https://react.dev/
3. **Tailwind CSS:** https://tailwindcss.com/docs
4. **Supabase (Backend & Auth):** https://supabase.com/docs
5. **Groq (AI Inference API):** https://console.groq.com/docs
6. **Recharts (Charting Library):** https://recharts.org/
7. **ShadCN/UI (UI Components):** https://ui.shadcn.com/
8. **Vercel (Deployment Platform):** https://vercel.com/docs

Glossary

Vite: The modern build tool and development server used to create the frontend application. Think of it as the high-speed workshop where the user interface is assembled and tested, making the development process incredibly fast.

React: The core JavaScript library used to build the dynamic and interactive user interface. It's like a set of digital LEGO bricks, allowing the creation of reusable components (buttons, charts, dialogs) that make up the entire application.

Supabase: The all-in-one backend platform responsible for the database, user authentication, and data security. It's the project's entire back office, providing a secure PostgreSQL database (the filing cabinet) and user authentication (the security desk) in one package.

Groq: The high-speed API used to run the AI models that power the application's intelligent features. Think of it as the super-fast engine that runs the AI's 'brain,' allowing it to think and respond almost instantly to user requests for advice or meal plans.

LLM (Large Language Model): The core AI "brain" (e.g., LLaMA 3 accessed via Groq) that powers the AI features. It's the expert nutritionist and fitness coach that has read countless books and can generate human-like text, analysis, and structured plans for all the diet and exercise tools.

AI Feature: A specialized program that uses the LLM to perform a single, focused task. Think of it as a highly-trained specialist; one feature is an expert on meal planning, another on exercise science, and another on answering nutrition questions.

Recharts: A specialized library used to render the interactive charts for features like weight tracking. It's the digital artist that turns raw data (like weight history) into easy-to-understand visual graphs.

Tailwind CSS & shadcn/ui: The frameworks used for styling and building the user interface components. This duo acts as the project's interior design and furniture kit. Tailwind provides the raw materials (colors, spacing), while shadcn/ui provides pre-designed, customizable furniture pieces (buttons, forms, cards) to build a beautiful and functional interface quickly.